



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Total Boundness and Compactness

ϵ -net and Total Boundness

APRIL 28, 2026

CONDUCTED BY

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ϵ -net and Total Boundness

ϵ -net

Let (X, d) be a metric space and $\epsilon > 0$. A subset $N \subseteq X$ is called an ϵ -net for X if for every point $x \in X$, there exists a point $n \in N$ such that $d(x, n) < \epsilon$. In other words, the collection of open balls $\{B(n, \epsilon) : n \in N\}$ covers the entire space X .

② Example of ϵ -net

Show that the set $Z = \{-3, -2, -1, 0, 1, 2, 3\}$ is a 1-net for $\mathbb{R}$.



② Example of ϵ -net

The set $B = \{B(m, n) : m, n \in \mathbb{Z}\}$ is a $\sqrt{2}$ -net for $\mathbb{R}^2$.



Total Boundness

A metric space (X, d) is said to be *totally bounded* if for every $\epsilon > 0$, there exists a finite ϵ -net for X . In other words, for any given $\epsilon > 0$, we can find a finite set of points in X such that the open balls centered at those points with radius ϵ cover the entire space X .

Example of Total Boundness

The interval $[0, 1]$ is totally bounded.



② Example of Total Boundedness

The set $\mathbb{R}$ is not totally bounded.



② Example of Total Boundedness

The infinite discrete space $\mathbb{N}$ with the discrete metric is not totally bounded.



② Example of Total Boundedness

The infinite-dimensional space ℓ^2 is not totally bounded.



Separability and Compactness

Totally Boundedness and Separability

Every totally bounded metric space is separable.



Compactness and Separability

Every compact metric space is separable.



Total Boundness and Compactness

Total Boundness and Cauchy Subsequences

A metric space (X, d) is totally bounded if and only if every sequence in X contains a Cauchy subsequence. 

Sequential Compactness and Total Boundness

A metric space (X, d) is sequentially compact if and only if it is complete and totally bounded. 

Lebesgue number and Lebesgue Covering Lemma

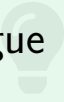
Lebesgue Number

Let $\{G_\alpha : \alpha \in \Lambda\}$ be an open cover of a metric space (X, d) . A real number $\lambda > 0$ is said to be a *Lebesgue number* for the open cover $\{G_\alpha\}$ if for each subset A of X with $d(A) < \lambda$, there is at least one set G_α which contains A .

In other words, any subset of X that is "small enough" (in terms of diameter) can be entirely contained within one of the sets in the cover $\mathcal{U}$.

Lebesgue Covering Lemma

Every open cover of a sequentially compact metric space has a Lebesgue number.



Sequential Compactness and Compactness

💡 In MS, Sequential Compactness $\iff$ Compactness

A metric space (X, d) is sequentially compact if and only if it is compact. 💡

Continuity $\iff$ Uniform Continuity on Compact Sets

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a continuous function. If X is compact, then f is uniformly continuous.

Equivalent Characterizations of Compactness

💡 Equivalent Characterizations of Compactness

For a metric space (X, d) , the following statements are equivalent: 

1. X is compact.
2. X is sequentially compact.
3. X is complete and totally bounded.
4. Bolzano-Weierstrass property

Thank You!

We'd love your questions and feedback.

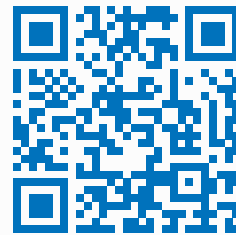
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(Lectures, walkthroughs, and course updates)



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References

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- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.